Задача 1.

1. 1.

$$\bar{f}_1(1, -3, 2), \bar{f}_2(3, 2, -4), \bar{f}_3(4, 0, -1)$$

$$\Gamma = \begin{pmatrix} (\bar{f}_1; \bar{f}_1) & (\bar{f}_1; \bar{f}_2) & (\bar{f}_1 \cdot \bar{f}_3) \\ (\bar{f}_2; \bar{f}_1) & (\bar{f}_2 \cdot \bar{f}_2) & (\bar{f}_2 \cdot \bar{f}_3) \\ (\bar{f}_3; \bar{f}_1) & (\bar{f}_3; \bar{f}_2) & (\bar{f}_3; \bar{f}_3) \end{pmatrix}$$

$$(\bar{f}_1; \bar{f}_1) = 1^2 + (-3)^2 + 2^2 = 14$$

$$(\bar{f}_1; \bar{f}_2) = 1 \cdot 3 - 3 \cdot 2 - 2 \cdot 4 = -11$$

$$(\bar{f}_1; \bar{f}_3) = 4 - 2 = 2$$

$$(\bar{f}_2; \bar{f}_2) = 9 + 4 + 16 = 29$$

$$(\bar{f}_3; \bar{f}_3) = 17$$

$$(\bar{f}_2; \bar{f}_3) = 16$$

$$(\bar{f}_1; \bar{f}_3) = 2$$

$$\Gamma = \begin{pmatrix} 14 & -11 & 2 \\ -11 & 29 & 16 \\ 2 & 16 & 17 \end{pmatrix} |\Gamma| = 14 \cdot 29 \cdot 17 - 22 \cdot 16 - 32 \cdot 11 -$$

$$- 29 \cdot 4 - 256 \cdot 14 - 121 \cdot 17 = 441$$

1.2. $\bar{f}_1(1;0;2;-1)$, $\bar{f}_2(2;1;0;3)$, $\bar{f}_3(-1;2;1;0)$; $\bar{f}_4(0;-1;3;2)$

$(\bar{f}_1;\bar{f}_1) = 1+4+1 = 6$

$(\bar{f}_1;\bar{f}_2) = 2-3 = -1$

$(\bar{f}_1;\bar{f}_3) = -1+2 = 1$

$(\bar{f}_1;\bar{f}_4) = 6-2 = 4$

$(\bar{f}_2;\bar{f}_3) = -2+2 = 0$

$(\bar{f}_2;\bar{f}_4) = -1+6 = 5$

$(\bar{f}_3;\bar{f}_4) = -2+3 = 1$

$(\bar{f}_3;\bar{f}_3) = 1+4+1 = 6$

$(\bar{f}_2;\bar{f}_2) = 4+1+9 = 14$

$(\bar{f}_4;\bar{f}_4) = 1+9+4 = 14$

$$\Gamma = \begin{pmatrix} 6 & -1 & 1 & 4 \\ -1 & 14 & 0 & 5 \\ 1 & 0 & 6 & 1 \\ 4 & 5 & 1 & 14 \end{pmatrix} \qquad |\Gamma| = 4356.$$

1.3. $\bar{f}_1(2;-1;3)$, $\bar{f}_2(0;4;-2)$

$(\bar{f}_1;\bar{f}_1) = 4+1+9 = 14$ $(\bar{f}_2;\bar{f}_2) = 16+4 = 20$

$(\bar{f}_1;\bar{f}_2) = -4-6 = -10$

$$\Gamma = \begin{pmatrix} 14 & -10 \\ -10 & 20 \end{pmatrix}$$

$|\Gamma| = 280 - 100 = 180$

2.1. Задача 2.

$$[\hat{A}]_{б} = \begin{pmatrix} -3 & 2 \\ 1 & -1 \end{pmatrix} \qquad T_{б \to б'} = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$[\hat{A}]_{б'} = T_{б' \to б} \cdot [\hat{A}]_{б} \cdot T_{б \to б'}$$

$$T_{б' \to б} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$$

$$[\hat{A}]_{б'} = \frac{1}{3} \cdot \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -3 & 2 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} =$$

$$= \frac{1}{3} \begin{pmatrix} -3+1 & 2-1 \\ 3+2 & -2-2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -2 & 1 \\ 5 & -4 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} =$$

$$= \frac{1}{3} \begin{pmatrix} -4+1 & 2+1 \\ 10-4 & -5-4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -3 & 3 \\ 6 & -9 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 2 & -3 \end{pmatrix}$$

2.2.

$$[\hat{A}]_{б} = \begin{pmatrix} 5 & -1 \\ 0 & 2 \end{pmatrix}$$

$$T_{б \to б'} = \begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix}$$

$$[\hat{A}]_{б'} = T_{б' \to б} \cdot [\hat{A}]_{б} \cdot T_{б \to б'}$$

$$T_{б' \to б} = \frac{1}{-5} \begin{pmatrix} -1 & -1 \\ -2 & 3 \end{pmatrix}$$

$$[\hat{A}]_{\sigma'} = \frac{1}{5}\begin{pmatrix} 1 & 1 \\ 2 & -3 \end{pmatrix} \cdot \begin{pmatrix} 5 & -1 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix} =$$

$$= \frac{1}{5} \cdot \begin{pmatrix} 5 & -1+2 \\ 10 & -2-6 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 5 & 1 \\ 10 & -8 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix} =$$

$$= \frac{1}{5}\begin{pmatrix} 15+2 & 5-1 \\ 30-16 & 10+8 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 17 & 4 \\ 14 & 18 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{17}{5} & \frac{4}{5} \\ \frac{14}{5} & \frac{18}{5} \end{pmatrix}$$

2.3. $[\hat{A}]_{\sigma} = \begin{pmatrix} 1 & 4 \\ -2 & 0 \end{pmatrix}$

$$T_{\sigma \to \sigma'} = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$$

$$T_{\sigma' \to \sigma} = T_{\sigma \to \sigma'}^{-1} = \frac{1}{-5}\begin{pmatrix} 1 & -2 \\ -3 & 1 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -1 & 2 \\ 3 & -1 \end{pmatrix}$$

$$[\hat{A}]_{\sigma'} = T_{\sigma' \to \sigma} \cdot [\hat{A}]_{\sigma} \cdot T_{\sigma \to \sigma'} = \frac{1}{5}\begin{pmatrix} -1 & 2 \\ 3 & -1 \end{pmatrix} \times$$

$$\times \begin{pmatrix} 1 & 4 \\ -2 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -1-4 & -4 \\ 3+2 & 12 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} =$$

$$= \frac{1}{5}\begin{pmatrix} -5 & -4 \\ 5 & 12 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -5-12 & -10-4 \\ 5+36 & 10+12 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{-17}{5} & \frac{-14}{5} \\ \frac{41}{5} & \frac{22}{5} \end{pmatrix}$$

Задача 3

3.1.

$$A = \begin{pmatrix} 9 & -8 & 4 \\ 8 & -7 & 4 \\ 4 & -4 & 3 \end{pmatrix}$$

$\underline{\lambda = 1}$:

$$\begin{pmatrix} 8 & -8 & 4 \\ 8 & -8 & 4 \\ 4 & -4 & 2 \end{pmatrix}_{III - 2\cdot II} \sim \begin{pmatrix} 8 & -8 & 4 \\ 8 & -8 & 4 \\ \cancel{0} & \cancel{0} & \cancel{0} \end{pmatrix}_{II - I} \sim \begin{pmatrix} 8 & -8 & 4 \\ \cancel{0} & \cancel{0} & \cancel{0} \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & \frac{1}{2} \end{pmatrix}$$

x_1 - связ

x_2, x_3 - своб.

$$x_1 - x_2 + \frac{1}{2} x_3 = 0$$

$$x_1 = x_2 - \frac{1}{2} x_3$$

ФСР: $\begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_{\lambda=1} = \alpha_1 \cdot \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} + \alpha_2 \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \alpha_1^2 + \alpha_2^2 > 0$$

3.2.

$$A = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$\underline{\lambda = 3}$:

$$\begin{pmatrix} 2 & 2 & 0 \\ 2 & -1 & 0 \\ \cancel{0} & \cancel{0} & \cancel{0} \end{pmatrix} \sim \begin{pmatrix} 2 & 2 & 0 \\ 2 & -1 & 0 \end{pmatrix}_{II-I} \sim \begin{pmatrix} 2 & 2 & 0 \\ 0 & -3 & 0 \end{pmatrix} \sim \begin{pmatrix} 2 & 2 & 0 \\ 0 & 2 & 0 \end{pmatrix}_{I-II} \sim$$

x_1, x_2 - связ

x_3 - своб.

$$\sim \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$x_1 = 0$

$x_2 = 0$

ФСР: $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_{\lambda=3} = \alpha \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \alpha \neq 0$$

3.3.

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$\underline{\lambda = 2}$:

$$\begin{pmatrix} 1 & 1 & -1 \\ \cancel{0} & \cancel{0} & \cancel{0} \\ 1 & 1 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix} \sim \begin{pmatrix} \overset{\downarrow}{1} & 1 & -1 \end{pmatrix}$$

x_1 - связ

x_2, x_3 - своб.

$$x_1 = x_3 - x_2$$

ФСР: $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_{\lambda = 2} = \alpha_1 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad \alpha_1^2 + \alpha_2^2 > 0$$

Задача 4.

4.1.

$$A = \begin{pmatrix} 1 & -2 & -3 & -4 \\ 6 & 1 & 0 & -1 \\ 4 & 5 & 6 & 7 \\ 9 & 8 & 9 & 11 \end{pmatrix} \underset{\substack{II - 6\cdot I \\ III - 4 I \\ IV - 9\cdot I}}{\sim} \begin{pmatrix} 1 & -2 & -3 & -4 \\ 0 & 13 & 18 & 23 \\ 0 & 13 & 18 & 23 \\ 0 & 26 & 36 & 47 \end{pmatrix}_{III - II} \sim$$

$$\sim \begin{pmatrix} 1 & -2 & -3 & -4 \\ 0 & 13 & 18 & 23 \\ \cancel{0} & \cancel{0} & \cancel{0} & \cancel{0} \\ 0 & 26 & 36 & 47 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -3 & -4 \\ 0 & 13 & 18 & 23 \\ 0 & 26 & 36 & 47 \end{pmatrix}_{III - 2\cdot II} \sim$$

$$\sim \begin{pmatrix} 1 & -2 & -3 & -4 \\ 0 & 13 & 18 & 23 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Rg(A) = 3$$

$$d(A) = \dim(V) - Rg(A) = 4 - 3 = 1$$

4.2.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix} \underset{\substack{II - 2\cdot I \\ III - 3\cdot I}}{\sim} \begin{pmatrix} 1 & 2 & 3 \\ \cancel{0} & \cancel{0} & \cancel{0} \\ \cancel{0} & \cancel{0} & \cancel{0} \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

x_1 - связ

x_2, x_3 - своб.

$$Rg(A) = 1$$

$$Rg(A) + d(A) = \dim(V) \Rightarrow d(A) = 3 - 1 = 2$$

4.3.

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 1 & 0 & 1 \\ 1 & 3 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 1 & 0 & 1 \\ 1 & 3 & 2 & 1 \end{pmatrix}_{\substack{II-2\cdot I \\ III-3\cdot I \\ IV-I}} \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ \cancel{0} & \cancel{0} & \cancel{0} & \cancel{0} \\ 0 & -5 & -9 & -11 \\ 0 & 1 & -1 & -3 \end{pmatrix}_{III\cdot(-1)} \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 9 & 11 \\ 0 & 1 & -1 & -3 \end{pmatrix}_{III-5\cdot II} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 9 & 11 \\ 0 & 0 & -46 & -58 \end{pmatrix}$$

x_1, x_2, x_3 - баз

x_4 - своб.

$$Rg(A) = 3$$

$$Rg(A) + d(A) = \dim(V) \Rightarrow d(A) = 4 - 3 = 1$$

Задача 5.

5.1. $f(x,y,z) = 4x^2 + 2y^2 + z^2 - 4xy - 2yz$

$$\begin{pmatrix} 4 & -2 & 0 \\ -2 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

$\Delta_1 = 4 > 0$

$\Delta_2 = 8 - 4 = 4 > 0$

$\Delta_3 = 8 - 4 - 4 = 0$

$2 > 0, \quad 1 > 0$

$$\begin{vmatrix} 4 & 0 \\ 0 & 1 \end{vmatrix} = 4 > 0$$

$$\begin{vmatrix} 2 & -1 \\ -1 & 1 \end{vmatrix} = 2 - 1 = 1 > 0$$

$\Rightarrow$ положительно полуопределённая, но не положительно определённая

5.2. $f(x,y,z) = 3x^2 + 5y^2 + 2z^2 + 2xy + 4xz$

$$\begin{pmatrix} 3 & 1 & 2 \\ 1 & 5 & 0 \\ 2 & 0 & 2 \end{pmatrix}$$

$\Delta_1 = 3 > 0$

$\Delta_2 = 15 - 1 = 14 > 0$

$\Delta_3 = 30 - 20 - 2 = 8 > 0$

$\Delta_1 > 0$, $\Delta_2 > 0$, $\Delta_3 > 0$ => положительно определённая

5.3. $f(x,y,z) = -2x^2 - 4y^2 - 5z^2 + 2xy + 4xz + 2yz$

$$\begin{pmatrix} -2 & 1 & 2 \\ 1 & -4 & 1 \\ 2 & 1 & -5 \end{pmatrix}$$

$\Delta_1 = -2 < 0$

$\Delta_2 = 8 - 1 = 7 > 0$

$\Delta_3 = -40 + 2 + 2 + 16 + 2 + 5 = -13 < 0$

$\Delta_1 < 0$, $\Delta_2 > 0$, $\Delta_3 < 0$ => отрицательно определённая

5.4. $f(x,y,z) = 5x^2 + 2y^2 - z^2 - 4xy - 2yz$

$$\begin{pmatrix} 5 & -2 & 0 \\ -2 & 2 & -1 \\ 0 & -1 & -1 \end{pmatrix}$$

$\Delta_1 = 5 > 0$

$\Delta_2 = 10 - 4 = 6 > 0$

$\Delta_3 = -10 - 5 - 4 = -19 < 0$

$\Delta_1 > 0,\ \Delta_2 > 0,\ \Delta_3 < 0 \Rightarrow$ знакопеременная

5.5. $f(x,y,z) = 2y^2 + z^2 - 4xy - 2yz$

$$\begin{pmatrix} 0 & -2 & 0 \\ -2 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

$\Delta_1 = 0 \leq 0$

$\Delta_2 = -4 < 0$

$\Delta_3 = -4 < 0$

$\Delta_1 \leq 0,\ \Delta_2 < 0,\ \Delta_3 < 0 \Rightarrow$ знакопеременная

5.6. $f(x,y,z) = -x^2 - y^2 - z^2 - 2xy - 2xz - 2yz$

$$\begin{pmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix}$$

$\Delta_1 = -1 < 0$

$\Delta_2 = 1 - 1 = 0$

$\Delta_3 = -1 - 1 - 1 + 1 + 1 + 1 = 0$

$$\begin{vmatrix} -1 & -1 \\ -1 & -1 \end{vmatrix} = 0 \qquad \begin{vmatrix} -1 & -1 \\ -1 & -1 \end{vmatrix} = 0$$

=> отрицательно полуопределённая, но не отрицательно определённая

5.7. $f(x, y, z) = 3x^2 - y^2 + 2z^2 + 4xy + 6xz$

$$\begin{pmatrix} 3 & 2 & 3 \\ 2 & -1 & 0 \\ 3 & 0 & 2 \end{pmatrix}$$

$\Delta_1 = 3 > 0$

$\Delta_2 = -3 - 4 = -7 < 0$

$\Delta_3 = -6 + 9 - 6 = -3 < 0$

$\Delta_1 > 0$, $\Delta_2 < 0$, $\Delta_3 < 0$ => знакопеременная

5.8. $f(x, y, z) = 4x^2 + y^2 + z^2 - 4xy + 4xz - 2yz$

$$\begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix}$$

$\Delta_1 = 4 > 0$

$\Delta_2 = 4 - 4 = 0$

$\Delta_3 = 4 + 4 + 4 - 4 - 4 - 4 = 0$

$1 > 0$, $1 > 0$

$$\begin{vmatrix} 4 & 2 \\ 2 & 1 \end{vmatrix} = 4 - 4 = 0 \qquad \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} = 1 - 1 = 0$$

=> положительно полуопределённая, но

не положительно определённая.

5.9. $f(x,y,z) = x^2 + z^2 + 4xz$

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

$\Delta_1 = 1$

$\Delta_2 = 0$

$\Delta_3 = 0 + 0 + 0 - 0 - 0 - 0 = 0$

$0 = 0, \quad 1 > 0$

$$\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 - 4 = -3$$

=> форма знакопеременная.

Задача 6.

6.1.

$$(x^2-y^2)\,dx+3xy\cos\left(\frac{y}{x}\right)dy=0$$

$M(x,y)dx+N(x,y)dy=0$ – ДУ с разделяющимися переменными.

$$M(x,y)=x^2-y^2;\quad N(x,y)=3xy\cos\left(\frac{y}{x}\right)$$

$$M(tx,ty)=t^2x^2-t^2y^2=t^2(x^2-y^2)=t^2\cdot M(x,y)$$

$$N(tx,ty)=3tx\,ty\cos\left(\frac{ty}{tx}\right)=t^2\cdot 3xy\cos\left(\frac{y}{x}\right)=$$

$$=t^2\cdot N(x,y)$$

Подстановка $y=zx$

$$(x^2-z^2x^2)\,dx+3xzx\cos\left(\frac{zx}{x}\right)\cdot(x\,dz+z\,dx)=0$$

$$(x^2-z^2x^2)\,dx+3x^2z^2\cos(z)\,dx+3x^3z\cos z\,dz=0$$

$$\left(x^2-z^2x^2+3x^2z^2\cos(z)\right)dx+3x^3z\cos z\,dz=0$$

$$\underbrace{x^2}_{\varphi_1(x)}\underbrace{\left(1-z^2+3z^2\cos(z)\right)}_{\psi_1(z)}dx+\underbrace{x^3}_{\varphi_2(x)}\cdot\underbrace{3z\cos z}_{\psi_2(z)}\,dz=0$$

6.2. $\quad y'+2xy=y^4\,\mathrm{tg}\,x$

$y'+a(x)\cdot y=f(x)\cdot y^{\alpha},\ \alpha\neq 0,\ \alpha\neq 1$ – ДУ Бернулли

$a(x) = 2x, \ f(x) = tg(x), \ \alpha = 4$

$y = uv$

$y' = u'v + v'u$

$u'v + v'u + 2x \cdot uv = tg(x) \cdot u^4 \cdot v^4$

$(u' + 2x \cdot u) \cdot v + v'u = tg(x) \cdot u^4 v^4$

Пусть $u' + 2xu = 0$

$\frac{du}{dx} = -2xu$

$du = -2xu\,dx$

$\frac{du}{u} + 2x\,dx = 0$

$\int \frac{du}{u} + 2\int x\,dx = 0$

$\ln|u| + x^2 = 0$

$\ln|u| = -x^2$

$|u| = e^{-x^2}$

$u = \pm e^{-x^2}$ - выберем +

$v' \cdot \cancel{e^{-x^2}} = tg(x) \cdot e^{-4x^2} v^4 \ \Big| : e^{-x^2}$

$$v' = tg(x)\cdot e^{-3x^2} v^4$$

$$dV = tg(x)\cdot e^{-3x^2}\cdot V^4 dx$$

$$\underbrace{V^{-4}}_{\psi_1(v)} dV - \underbrace{tg(x)\cdot e^{-3x^2}}_{\varphi_2(x)} dx = 0$$

$\varphi_1(x) \equiv 1$ $\quad \psi_2(v) \equiv 1$

6.3. $(y^2-2x^2)dx + 2xy\ln\left(\frac{y}{x}\right)dy = 0$

$$M(x,y)dx + N(x,y)dy = 0$$

$$M(x,y) = y^2-2x^2;\ N(x,y) = 2xy\ln\left(\frac{y}{x}\right)$$

$$M(tx,ty) = t^2y^2-2t^2x^2;\ N(tx,ty) = 2tx\,ty\cdot\ln\left(\frac{ty}{tx}\right)$$

$$t^2(y^2-2x^2) + t^2\cdot\left(2xy\ln\left(\frac{y}{x}\right)\right) = 0$$

$$t^2\cdot M(x,y) + t^2\cdot N(x,y) = 0$$

Это однородное ДУ степени однородности

$K=2$.

Подстановка $y = zx$:

$$(z^2x^2-2x^2)dx + 2x^2z\ln\left(\frac{z\cancel{x}}{\cancel{x}}\right)\cdot(zdx+xdz)$$

$$(z^2x^2-2x^2)dx + 2x^2z\ln(z)(zdx+xdz) = 0$$

$$\left(z^2x^2-2x^2+2x^2z^2\ln(z)\right)dx+2x^3z\ln(z)\,dz=0$$

$$\underbrace{x^2}_{\varphi_1(x)}\underbrace{\left(z^2-2+2z^2\ln(z)\right)}_{\psi_1(z)}dx+\underbrace{2x^3}_{\varphi_2(x)}\underbrace{z\ln(z)}_{\psi_2(z)}dz=0$$

6.4. $(x^2+y^2)\,dx-xy\sin\left(\frac{y}{x}\right)dy=0$

$$M(x,y)\,dx+N(x,y)\,dy=0$$

$$M(x,y)=x^2+y^2;\ N(x;y)=-xy\sin\left(\frac{y}{x}\right)$$

$$M(tx,ty)=t^2x^2+t^2y^2,\ N(tx,ty)=-tx\cdot ty\sin\left(\frac{ty}{tx}\right)$$

$$(t^2x^2+t^2y^2)\,dx-tx\,ty\sin\left(\frac{y}{x}\right)dy=0$$

$$t^2\cdot(x^2+y^2)\,dx-t^2xy\cdot\sin\left(\frac{y}{x}\right)dy=0$$

$$t^2\cdot M(x,y)\,dx-t^2\cdot N(x,y)\,dy=0$$

Это однородное ДУ-I.

Подстановка $y=zx$

$$(x^2+z^2x^2)\,dx-x^2z\sin(z)(z\,dx+x\,dz)=0$$

$$\left(x^2+z^2x^2-x^2z^2\sin(z)\right)dx-x^3z\sin(z)\,dz=0$$

$$\underbrace{x^2}_{\varphi_1(x)}\underbrace{\left(1+z^2-z^2\sin(z)\right)}_{\psi_1(z)}dx\ \underbrace{-x^3}_{\varphi_2(x)}\underbrace{z\sin(z)}_{\psi_2(z)}dz=0$$

6.5. $y' + 3x^2 y = y^3 \cos x$

$y' + a(x) y = f(x) \cdot y^{\alpha}, \alpha \neq 0, \alpha \neq 1$ - ДУ - Бернулли

$a(x) = 3x^2, f(x) = \cos(x), \alpha = 3.$

$y = uv$

$y' = u'v + v'u$

$u'v + v'u + 3x^2 \cdot uv = \cos(x) \cdot u^3 v^3$

$(u' + 3x^2 u) \cdot v + v'u = \cos(x) \cdot u^3 v^3$

Положим $u' + 3x^2 u = 0$

$\frac{du}{dx} = -3x^2 u \qquad du = -3x^2 u\, dx$

$\frac{du}{u} = -3x^2 dx$

$\int \frac{du}{u} = -3\int x^2 dx$

$\ln|u| = -x^3$

$u = \pm e^{-x^3}$ - выберем „+"

$v' \cdot \cancel{e^{-x^3}} = \cos(x) \cdot e^{-3x^3} \cdot v^3$

$$V' = \cos(x) \cdot e^{-2x^3} \cdot V^3$$

$$\frac{dV}{dx} = \cos(x) \cdot e^{-2x^3} \cdot V^3$$

$$dV = \cos(x) \cdot e^{-2x^3} \cdot V^3 dx \,\Big|\, : V^3$$

$$\underbrace{V^{-3}}_{\psi_1(V)} \underset{\varphi_1(x)\equiv 1}{dV} - \underbrace{\cos(x) \cdot e^{-2x^3}}_{\varphi_2(x)} \underset{\psi_2(V)\equiv 1}{dx} = 0$$

6.6. $$y' - \frac{y}{x} = y^2 \ln x$$

$$y' - a(x) \cdot y = f(x) \cdot y^{\alpha} \text{ — ДУ-Бернулли}$$

$$a(x) = x^{-1}, \quad f(x) = \ln(x), \quad \alpha = 2$$

$$y = uV$$

$$y' = u'V + V'u$$

$$u'V + V'u - \frac{uV}{x} = u^2 V^2 \cdot \ln(x)$$

$$\left(u' - \frac{u}{x}\right) \cdot V + V'u = u^2 V^2 \cdot \ln(x)$$

Положим $u' - \frac{u}{x} = 0$

$$\frac{du}{dx} = \frac{u}{x} \qquad du = \frac{u\,dx}{x} \,\Big|\, : u$$

$$\frac{du}{u} = \frac{dx}{x}$$

$$\int \frac{du}{u} = \int \frac{dx}{x}$$

$$\ln|u| = \ln|x|$$

$$|u| = |x|$$

$$\underline{u = x}$$

$$v' \cdot x = x^2 v^2 \cdot \ln(x)$$

$$\frac{dv}{dx} \cdot x = x^2 \cdot v^2 \ln x$$

$$x\,dv = x^2 v^2 \ln x\,dx$$

$$\underbrace{x^2 \ln x}_{\varphi_1(x)} \cdot \underbrace{v^2}_{\psi_1(v)} dx - \underbrace{x}_{\varphi_2(x)} dv = 0 \quad \psi_2(v) \equiv 0$$

Задача 7

7.1. $$\frac{yy''-(y')^2}{y^2}=4x+yy'$$

$$\frac{yy''-(y')^2}{y^2}=\left(\frac{y'}{y}\right)'$$

$$\left(\frac{y'}{y}\right)'-yy'=4x$$

$$yy'=\left(\frac{y^2}{2}\right)'$$

$$\left(\frac{y'}{y}\right)'-\left(\frac{y^2}{2}\right)'=4x$$

$$\left(\frac{y'}{y}-\frac{y^2}{2}\right)'=4x$$

$$\frac{y'}{y}-\frac{y^2}{2}=2x^2+C$$

$$\frac{y'}{y}=2x^2+\frac{y^2}{2}+C$$

7.2. $$yy'+yy''\ln(x)=x(y')^2$$

$$yy'+yy''\ln(x)-x(y')^2=0$$

$$F(x,y,y',y'')=yy'+yy''\ln(x)-x(y')^2$$

$$F(x,ty,ty',ty'')=ty\cdot ty'+ty\cdot ty''\ln(x)-x\cdot t^2(y')^2=$$

$= t^2 \cdot yy' + t^2 \cdot y \cdot y'' \ln(x) - t^2 \cdot x \cdot (y')^2 =$

$= t^2 \cdot (yy' + yy'' \ln(x) - x \cdot (y')^2) = t^2 \cdot F(x, y, y', y'')$

$\Rightarrow$ F - ф-ция одноp. отн. y, y', y'' (2-й степени) $\Rightarrow$ данное ДУ - однородное отн. ф-ции и её производных.

$y' = z(x) \cdot y \qquad y'' = (y')' = (z(x) \cdot y)' = z' \cdot y +$

$+ y' z = z' \cdot y + z^2 \cdot y = (z' + z^2) \cdot y$

$y \cdot z \cdot y + y \cdot (z' + z^2) \cdot y \ln(x) = x \cdot z^2 y^2$

$\cancel{y^2} \cdot z + \cancel{y^2} \cdot (z' + z^2) \cdot \ln(x) = x z^2 \cancel{y^2}$

$z + (z' + z^2) \cdot \ln(x) = x z^2$

$z' \ln(x) + z - x z^2 + z^2 \ln(x) = 0$

$z' \ln(x) + z + z^2 (\ln(x) - x) = 0$

7.3. $\quad 2xy' + x^2 y'' = 0$

$2xy' + x^2 y'' = (x^2 \cdot y')'$

$(x^2 \cdot y')' = 0$

$\int x^2 \cdot y' dx = \int 0 \, dx$

$x^2 \cdot y' = C$

$y' = \frac{C}{x}$

7.4. $y^2 y'' + 2(y')^2 y - 3y^2 y' = 0$

$F(x, y, y', y'') = y^2 y'' + 2(y')^2 y - 3y^2 y'$

$F(x, ty, ty', ty'') = t^2 y^2 \cdot ty'' + 2t^2 (y')^2 \cdot ty -$

$- 3t^2 y^2 \cdot ty' = t^3 \cdot y^2 \cdot y'' + 2t^3 \cdot (y')^2 \cdot y - 3t^3 \cdot y^2 \cdot y' =$

$= t^3 (y^2 \cdot y'' + 2y \cdot (y')^2 - 3y^2 \cdot y') = t^3 \cdot F(x, y, y', y'')$

$\Rightarrow F$-ф-ция однородная отн. y, y', y'' (3-й степени) $\Rightarrow$ данное ДУ-II однородно отн. неизв. ф-ции и её производных.

Подстановка $y' = z(x) \cdot y$

$y'' = z' \cdot y + y' z = z' \cdot y + z \cdot y \cdot z = (z' + z^2) \cdot y$

$\cancel{y^3} \cdot (z' + z^2) + 2 \cdot z^2 \cancel{y^3} - 3\cancel{y^2} \cdot z\cancel{y} = 0$

$z' + z^2 + 2z^2 - 3z = 0$

$z' + 3z^2 - 3z = 0$

7.5. $y'' = xy' + y + 1$

$y'' = (y')'$

$xy' + y = (xy)'$

$x' = 1$

$(y')' = (xy)' + (x)'$

$(y')' = (xy + x)'$

$y' = xy + x + C_1$

7.6. $xy'' = 2yy' - y'$

$xy'' + y' = 2yy'$

$(xy')' = 2yy'$

$(y^2)' = 2 \cdot y \cdot y'$

$(xy')' = (y^2)'$

$xy' = y^2 + C$

Задача 8.

8.1.

$$y'' - 4y' + 4y = (4x+1)\cos 2x + 5\sin 2x$$

Соотв. ОЛ ДУ:

$$y'' - 4y' + 4y = 0$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$(\lambda - 2)^2 = 0$$

$$\lambda_{1,2} = 2$$

ФСР: $e^{2x}, \; xe^{2x}$ $\qquad y_{о.о.} = C_1 \cdot e^{2x} + C_2 x e^{2x}$

Возвр. к неодн. ДУ:

$$F(x) = (4x+1)\cos 2x + 5\sin 2x = e^{0 \cdot x}\big((4x+1)\cdot\cos 2x + 1 \times$$

$$\times 5\sin 2x\big)$$

$$\alpha = 0, \; \beta = 2, \; P(x) = 4x+1, \; Q(x) = 1$$

$$\alpha + \beta i = 2i \Rightarrow S = 0$$

$$\deg R = \deg T = \max\{\overset{=1}{\deg}(P), \deg(\overset{=0}{Q})\} = 1$$

$$R(x) = A + Bx, \; T(x) = C + Dx.$$

$$y_{ч.н.} = x^0 \cdot e^0\big((A+Bx)\cos 2x + (C+Dx)\sin 2x\big) =$$

$$= \big((A+Bx)\cos 2x + (C+Dx)\sin 2x\big)$$

$y_{о.н.} = y_{о.о.} + y_{ч.н}$

$y_{о.н.} = C_1 \cdot e^{2x} + C_2 \cdot xe^{2x} + \Big((A+Bx)\cos 2x +$

$+ (C+Dx)\sin 2x\Big)$, где A, B, C, D - конкретные,

C_1, C_2 - любые

8.2. $y'' + 4y = 3x\cos(2x)$

Соотв. ОЛДУ:

$y'' + 4y = 0$

$\lambda^2 + 4 = 0$

$\lambda_{1,2} = \pm 2i$

ФСР: $\cos(2x)$, $\sin(2x)$

$y_{о.о.} = C_1 \cdot \cos(2x) + C_2 \cdot \sin(2x)$

Возвр. к нел ДУ:

$F(x) = 3x\cos(2x) = e^{0\cdot x}(3x\cos(2x) + 0\cdot\sin(2x))$

$\alpha = 0$, $\beta = 2$, $P(x) = 3x$, $Q(x) = 0$

$\alpha + \beta i = 0 + 2i = 2i \Rightarrow S = 1$

$\deg(R) = \deg(T) = \max\{\overset{=1}{\deg(P)}, \overset{=0}{\deg(Q)}\} =$

$= 1$

$R(x) = A + Bx$, $T(x) = C + D(x)$

$$y_{о.н.} = x^1 \cdot e^{0x}\left((A+Bx)\cos(2x) + (C+D(x))\sin(2x)\right) =$$

$$= (A+Bx) \cdot x\cos(2x) + (C+Dx) \cdot x\sin 2x$$

$$y_{о.н.} = y_{о.о.} + y_{ч.н.}$$

$$y_{о.н.} = C_1 \cdot \cos(2x) + C_2 \cdot \sin(2x) + x \cdot \left((A+Bx)\cos(2x) +\right.$$

$$\left. + (C+Dx)\sin 2x\right),$$ где C_1, C_2 - конкретные,

A, B, C, D - любые

Задача 9.

9.1.

$$\begin{cases} \dot{x} = x + y \\ \dot{y} = 3y - 2x \end{cases}$$

$$\begin{pmatrix} 1 & 1 \\ -2 & 3 \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & 1 \\ -2 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & 1 \\ -2 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) + 2 = 3 - 3\lambda - \lambda + \lambda^2 + 2 =$$

$$= \lambda^2 - 4\lambda + 5$$

$$\lambda^2 - 4\lambda + 5 = 0$$

$$D = 16 - 20 = -4$$

$$\lambda_{1,2} = \frac{4 \pm \sqrt{4}\, i}{2} = 2 \pm i$$

$\underline{\lambda = 2 + i}$:

$$\begin{pmatrix} -1-i & 1 \\ -1 & 1-i \end{pmatrix} \underset{\text{II}\cdot(1+i)}{\sim} \begin{pmatrix} -1-i & 1 \\ -1-i & 1 \end{pmatrix} \underset{\text{II}-\text{I}}{\sim} \begin{pmatrix} -1-i & 1 \\ 0 & 0 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} -1-i & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -\frac{1}{1+i} \end{pmatrix}$$

$$x_1 = \frac{1}{1+i} x_2$$

ФСР: $\begin{pmatrix} 1 \\ 1+i \end{pmatrix}$

$$\begin{pmatrix} 1 \\ 1+i \end{pmatrix} \cdot e^{(2+i)t} = \begin{pmatrix} 1 \\ 1+i \end{pmatrix} \cdot e^{2t} \cdot e^{it} =$$

$$= \begin{pmatrix} 1 \\ 1+i \end{pmatrix} \cdot e^{2t} \cdot \left(\cos(t) + i\sin(t) \right) =$$

$$= e^{2c} \cdot \begin{pmatrix} \cos(t) + i\sin(t) \\ (1+i)\cos(t) + (1+i)\cdot i\sin(t) \end{pmatrix} =$$

$$= e^{2t} \cdot \begin{pmatrix} \cos(t) + i\sin(t) \\ \cos(t) + i\cos(t) + i\sin(t) - \sin(t) \end{pmatrix} =$$

$$= e^{2t} \cdot \begin{pmatrix} \cos(t) \\ \cos(t) - \sin(t) \end{pmatrix} + e^{2t} \cdot i \cdot \begin{pmatrix} \sin(t) \\ \cos(t) + \sin(t) \end{pmatrix}$$

ФСР СОЛДУ:

$$e^{2t} \begin{pmatrix} \cos(t) \\ \cos(t) - \sin(t) \end{pmatrix}, \quad e^{2t} \begin{pmatrix} \sin(t) \\ \cos(t) + \sin(t) \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{o.o.} = C_1 \cdot e^{2t} \begin{pmatrix} \cos t \\ \cos t - \sin t \end{pmatrix} + C_2 \cdot e^{2t} \begin{pmatrix} \sin t \\ \cos t + \sin t \end{pmatrix}$$

$\times i$.

9.2.

$$\begin{cases} \dot{x} = 6x + y \\ \dot{y} = 5x + 2y \end{cases}$$

$$\begin{pmatrix} 6 & 1 \\ 5 & 2 \end{pmatrix}$$

$$\begin{vmatrix} 6-\lambda & 1 \\ 5 & 2-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 6-\lambda & 1 \\ 5 & 2-\lambda \end{vmatrix} = (6-\lambda)(2-\lambda) - 5 = 12 - 2\lambda - 6\lambda + \lambda^2 -$$

$$-5 = \lambda^2 - 8\lambda + 7$$

$$\lambda^2 - 8\lambda + 7 = 0$$

$$D = 64 - 28 = 36$$

$$\lambda_{1,2} = \frac{8 \pm 6}{2} = 7;\ 1$$

$\underline{\lambda = 1:}$

$$\begin{pmatrix} 5 & 1 \\ 5 & 1 \end{pmatrix}_{II-I} \sim \begin{pmatrix} 5 & 1 \\ \cancel{0} & \cancel{0} \end{pmatrix} \sim \begin{pmatrix} 5 & 1 \end{pmatrix}_{I:5} \sim \begin{pmatrix} 1 & \frac{1}{5} \end{pmatrix}$$

$$x_1 + \frac{1}{5} x_2 = 0$$

$$x_1 = -\frac{1}{5} x_2$$

ФСР: $\begin{pmatrix} 1 \\ -5 \end{pmatrix}$

$\underline{\lambda = 7:}$

$$\begin{pmatrix} -1 & 1 \\ 5 & -5 \end{pmatrix}_{II \cdot (-\frac{1}{5})} \sim \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}_{II-I} \sim \begin{pmatrix} -1 & 1 \\ \cancel{0} & \cancel{0} \end{pmatrix} \sim \begin{pmatrix} -1 & 1 \end{pmatrix}_{I \cdot (-1)} \sim$$

$\sim \begin{pmatrix} 1 & -1 \end{pmatrix}$

$x_1 - x_2 = 0$

$x_1 = x_2$

ФСР: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

ФСР СОЛДУ:

$$\begin{pmatrix} x \\ y \end{pmatrix}_{о.о.} = C_1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot e^{7t} + C_2 \cdot \begin{pmatrix} 1 \\ -5 \end{pmatrix} \cdot e^{t}$$

9.3. $\begin{cases} \dot{x} = 2x - 5y \\ \dot{y} = x - 2y \end{cases}$

$$\begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix}$$

$$\begin{vmatrix} 2-\lambda & -5 \\ 1 & -2-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2-\lambda & -5 \\ 1 & -2-\lambda \end{vmatrix} = (2-\lambda)(-2-\lambda) + 5 =$$

$$= -4 + 2\lambda + \lambda^2 - 2\lambda + 5 = \lambda^2 + 1$$

$\lambda^2 + 1 = 0$

$\lambda_{1,2} = \pm i$

$\underline{\lambda = i}$:

$$\begin{pmatrix} 2-i & -5 \\ 1 & -2-i \end{pmatrix}_{\mathrm{II}\cdot(2-i)} \sim \begin{pmatrix} 2-i & -5 \\ 2-i & -5 \end{pmatrix}_{\mathrm{II}-\mathrm{I}} \sim \begin{pmatrix} 2-i & -5 \\ \cancel{0} & \cancel{0} \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 2-i & -5 \end{pmatrix}_{\mathrm{I}:(2-i)} \sim \begin{pmatrix} 1 & -\frac{5}{2-i} \end{pmatrix}$$

$$x_1 = \frac{5}{2-i} x_2 = \frac{5(2+i)}{4+1} x_2 = (2+i)x_2$$

ФСР: $\begin{pmatrix} 2+i \\ 1 \end{pmatrix}$

$$\begin{pmatrix} 2+i \\ 1 \end{pmatrix} \cdot e^{it} = \begin{pmatrix} 2+i \\ 1 \end{pmatrix} \cdot (\cos(t) + i\sin(t)) =$$

$$= \begin{pmatrix} 2\cos(t) + i\cos(t) + 2i\sin(t) - \sin(t) \\ \cos(t) + i\sin(t) \end{pmatrix} =$$

$$= \begin{pmatrix} 2\cos(t) - \sin(t) \\ \cos(t) \end{pmatrix} + \begin{pmatrix} 2\sin(t) + \cos(t) \\ \sin(t) \end{pmatrix} \cdot i$$

ФСР соиДу:

$$\begin{pmatrix} 2\cos(t) - \sin(t) \\ \cos(t) \end{pmatrix}, \begin{pmatrix} 2\sin(t) + \cos(t) \\ \sin(t) \end{pmatrix} \cdot i$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{o.o.} = C_1 \cdot \begin{pmatrix} 2\cos(t) - \sin(t) \\ \cos(t) \end{pmatrix} + C_2 \begin{pmatrix} 2\sin(t) + \cos(t) \\ \sin(t) \end{pmatrix}$$

9.4.

$$\begin{cases} \dot{x} = x + 3y \\ \dot{y} = 2x + 2y \end{cases}$$

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & 3 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & 3 \\ 2 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda) - 6 = 2 - 2\lambda - \lambda + \lambda^2 -$$

$$-6 = \lambda^2 - 3\lambda - 4$$

$$\lambda^2 - 3\lambda - 4 = 0$$

$$D = 9 + 16 = 25$$

$$\lambda_{1,2} = \frac{3 \pm 5}{2} = 4; -1$$

$\underline{\lambda = -1}$:

$$\begin{pmatrix} 2 & 3 \\ 2 & 3 \end{pmatrix}_{\mathrm{II}-\mathrm{I}} \sim \begin{pmatrix} 2 & 3 \\ \cancel{0} & \cancel{0} \end{pmatrix} \sim \begin{pmatrix} 2 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & \frac{3}{2} \end{pmatrix}$$

$$x_1 + \frac{3}{2} x_2 = 0$$

$$x_1 = -\frac{3}{2} x_2$$

ФСР: $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

$\underline{\lambda = 4}$:

$$\begin{pmatrix} -3 & 3 \\ 2 & -2 \end{pmatrix}_{\substack{\mathrm{I} : (-3) \\ \mathrm{II} : 2}} \sim \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}_{\mathrm{II}-\mathrm{I}} \sim \begin{pmatrix} 1 & -1 \\ \cancel{0} & \cancel{0} \end{pmatrix} \sim \begin{pmatrix} 1 & -1 \end{pmatrix}$$

$x_1 - x_2 = 0$

$x_1 = x_2$

ФСР: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

ФСР СОЛДУ: $\begin{pmatrix} -3 \\ 2 \end{pmatrix} e^{-t}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t}$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{o.o.} = C_1 \cdot \begin{pmatrix} -3 \\ 2 \end{pmatrix} \cdot e^{-t} + C_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot e^{4t}$$

9.5. $\begin{cases} \dot{x} = 8y - x \\ \dot{y} = -x - y \end{cases}$

$$\begin{pmatrix} -1 & 8 \\ -1 & -1 \end{pmatrix}$$

$$\begin{vmatrix} -1-\lambda & 8 \\ -1 & -1-\lambda \end{vmatrix} = (1+\lambda)^2 + 8 = 1 + 2\lambda + \lambda^2 + 8 = \lambda^2 + 2\lambda + 9$$

$(1+\lambda)^2 = -8$

$1 + \lambda_{1,2} = \pm 2\sqrt{2}\, i$

$\lambda_{1,2} = -1 \pm 2\sqrt{2}\, i$

$\underline{\lambda = -1 + 2\sqrt{2}\, i:}$

$$\begin{pmatrix} 2\sqrt{2}i & 8 \\ -1 & -2\sqrt{2}i \end{pmatrix}_{\text{II}\cdot(2\sqrt{2}i)} \sim \begin{pmatrix} 2\sqrt{2}i & 8 \\ 2\sqrt{2}i & 8 \end{pmatrix}_{\text{II}-\text{I}} \sim \begin{pmatrix} 2\sqrt{2}i & 8 \\ \cancel{0} & \cancel{0} \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & \frac{4}{\sqrt{2}i} \end{pmatrix} \sim \begin{pmatrix} 1 & -2\sqrt{2}i \end{pmatrix}$$

ФСР: $\begin{pmatrix} 2\sqrt{2} \\ i \end{pmatrix}$

$$\begin{pmatrix} \sqrt{8} \\ i \end{pmatrix} \cdot e^{(-1+\sqrt{8}i)t} = \begin{pmatrix} \sqrt{8} \\ i \end{pmatrix} \cdot e^{-t} \cdot e^{\sqrt{8}it} =$$

$$= e^{-t} \cdot \begin{pmatrix} \sqrt{8} \\ i \end{pmatrix} \cdot \left(\cos(\sqrt{8}t) + i\sin(\sqrt{8}t)\right) =$$

$$= e^{-t} \cdot \begin{pmatrix} \sqrt{8}\cos(\sqrt{8}t) + \sqrt{8}i\sin(\sqrt{8}t) \\ i\cos(\sqrt{8}t) - \sin(\sqrt{8}t) \end{pmatrix} =$$

$$= e^{-t} \cdot \begin{pmatrix} \sqrt{8}\cos(\sqrt{8}t) \\ -\sin(\sqrt{8}t) \end{pmatrix} + e^{-t} \cdot i \cdot \begin{pmatrix} \sqrt{8}\sin(\sqrt{8}t) \\ \cos(\sqrt{8}t) \end{pmatrix}$$

ФСР сол ду:

$$C_1 \cdot e^{-t} \cdot \begin{pmatrix} \sqrt{8}\cos(\sqrt{8}t) \\ -\sin(\sqrt{8}t) \end{pmatrix}, \quad C_2 \cdot e^{-t} \cdot \begin{pmatrix} \sqrt{8}\sin(\sqrt{8}t) \\ \cos(\sqrt{8}t) \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{o.o.} = C_1 \cdot e^{-t} \begin{pmatrix} \sqrt{8}\cos(\sqrt{8}t) \\ -\sin(\sqrt{8}t) \end{pmatrix} + C_2 \cdot e^{-t} \begin{pmatrix} \sqrt{8}\sin(\sqrt{8}t) \\ \cos(\sqrt{8}t) \end{pmatrix}$$

9.6.

$$\begin{cases} \dot{x} = x - 8y \\ \dot{y} = -x - y \end{cases}$$

$$\begin{pmatrix} 1 & -8 \\ -1 & -1 \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & -8 \\ -1 & -1-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & -8 \\ -1 & -1-\lambda \end{vmatrix} = (1-\lambda)(-1-\lambda) - 8 =$$

$$= -1+\lambda - \lambda + \lambda^2 - 8 = \lambda^2 - 9$$

$$\lambda^2 - 9 = 0$$

$$\lambda_{1,2} = \pm 3$$

$\underline{\lambda = 3}$:

$$\begin{pmatrix} -2 & -8 \\ -1 & -4 \end{pmatrix} \sim \begin{pmatrix} -2 & -8 \\ -2 & -8 \end{pmatrix} \sim \begin{pmatrix} -2 & -8 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & 4 \end{pmatrix}$$

ФСР: $\begin{pmatrix} -4 \\ 1 \end{pmatrix}$

$\underline{\lambda = -3}$:

$$\begin{pmatrix} 4 & -8 \\ -1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 \\ 4 & -8 \end{pmatrix} \underset{II - 4 \cdot I}{\sim} \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 \end{pmatrix}$$

ФСР: $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

ФСР СОЛДУ:

$$e^{3t} \cdot \begin{pmatrix} -4 \\ 1 \end{pmatrix}, \quad e^{-3t} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{0.0.} = C_1 \cdot e^{3t} \begin{pmatrix} -4 \\ 1 \end{pmatrix} + C_2 \cdot e^{-3t} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Задача 10

10.1.

$$\begin{cases} \dot{x} = 2x - y \\ \dot{y} = 4x + 6y \end{cases}$$

$$y = 2x - \dot{x}$$

$$\dot{y} = 2\dot{x} - \ddot{x}$$

$$2\dot{x} - \ddot{x} = 4x + 6(2x - \dot{x})$$

$$4x + 12x - 6\dot{x} = 2\dot{x} - \ddot{x}$$

$$\ddot{x} - 8\dot{x} + 16x = 0$$

$$\lambda^2 - 8\lambda + 16 = 0$$

$$D = 64 - 64 = 0$$

$$\lambda_{1,2} = 4$$

$$\text{ФСР}: e^{4t}, te^{4t}$$

$$x_{o.o.} = C_1(t) \cdot e^{4t} + C_2(t) \cdot te^{4t}$$

$$y_{o.o.} = 2x_{o.o.} - \dot{x}_{o.o.}$$

$$y_{o.o.} = 2(C_1(t) \cdot e^{4t} + C_2(t) \cdot te^{4t}) -$$

$$-(4C_1(t) \cdot e^{4t} + 4C_2(t) \cdot te^{4t} + C_2(t) \cdot e^{4t}) =$$

$= 2C_1(t)\cdot e^{4t} + 2C_2(t)\cdot te^{4t} -$

$-4C_1(t)\cdot e^{4t} - 4C_2(t)\cdot te^{4t} - C_2(t)\cdot e^{4t} =$

$= -2C_1(t)e^{4t} - 2C_2(t)\cdot te^{4t} - C_2(t)\cdot e^{4t}$

$$\begin{cases} x_{o.o.} = C_1(t)\cdot e^{4t} + C_2(t)\cdot te^{4t} \\ y_{o.o.} = -2C_1(t)\cdot e^{4t} - 2C_2(t)\cdot te^{4t} - C_2(t)\cdot e^{4t} \end{cases}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{o.o.} = C_1\cdot\begin{pmatrix} 1 \\ -2 \end{pmatrix}\cdot e^{4t} + C_2\cdot\begin{pmatrix} t \\ -2t \end{pmatrix}\cdot e^{4t} +$$

$$+ C_2\begin{pmatrix} 0 \\ -1 \end{pmatrix}\cdot e^{4t} = C_1\cdot\begin{pmatrix} 1 \\ -2 \end{pmatrix}\cdot e^{4t} + C_2\begin{pmatrix} t \\ -1-2t \end{pmatrix}\cdot e^{4t}$$

10.2.

$$\begin{cases} \dot{x} = 5x - y \\ \dot{y} = x + 3y \end{cases}$$

$y = 5x - \dot{x}$

$\dot{y} = 5\dot{x} - \ddot{x}$

$5\dot{x} - \ddot{x} = x + 3(5x - \dot{x})$

$5\dot{x} - \ddot{x} = x + 15x - 3\dot{x}$

$\ddot{x} - 8\dot{x} + 16x = 0$

$\lambda^2 - 8\lambda + 16 = 0$

$(\lambda - 4)^2 = 0$

$\lambda_{1,2} = 4$

ФСР: e^{4t}, te^{4t}

$x_{o.o.} = C_1 \cdot e^{4t} + C_2 te^{4t}$

$y_{o.o.} = 5x_{o.o.} - \dot{x}_{o.o.} = 5(C_1 \cdot e^{4t} + C_2 \cdot te^{4t}) -$

$- 4C_2 te^{4t} - C_2 \cdot e^{4t} = 5C_1 e^{4t} + 5C_2 te^{4t} -$

$- 4C_2 te^{4t} - C_2 \cdot e^{4t} = 5C_1 e^{4t} + C_2 te^{4t} - C_2 e^{4t}$

$$\begin{cases} x_{o.o.} = C_1 \cdot e^{4t} + C_2 te^{4t} \\ y_{o.o.} = 5C_1 e^{4t} + (t-1) C_2 e^{4t} \end{cases}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{o.o.} = C_1 \cdot \begin{pmatrix} 1 \\ 5 \end{pmatrix} \cdot e^{4t} + C_2 \cdot \begin{pmatrix} t \\ t-1 \end{pmatrix} \cdot e^{4t}$$

10.3.

$$\begin{cases} \dot{x} = -2x - 5y \\ \dot{y} = 2x + 2y \end{cases}$$

$5y = -2x - \dot{x}$

$y = -\frac{2}{5}x - \frac{1}{5}\dot{x}$

$\dot{y} = -\frac{2}{5}\dot{x} - \frac{1}{5}\ddot{x}$

$-\frac{2}{5}\dot{x} - \frac{1}{5}\ddot{x} = 2x + 2\left(-\frac{2}{5}x - \frac{1}{5}\dot{x}\right)$

$\cancel{-\frac{2}{5}\dot{x}} - \frac{1}{5}\ddot{x} = 2x - \frac{4}{5}x \cancel{- \frac{2}{5}\dot{x}}$

$\frac{1}{5}\ddot{x} + \frac{6}{5}x = 0 \quad | \cdot 5$

$$\ddot{x} + 6x = 0$$

$$\lambda^2 + 6 = 0$$

$$\lambda_{1,2} = \pm\sqrt{6}\, i$$

ФСР: $\cos(\sqrt{6}t)$, $\sin(\sqrt{6}t)$

$$x_{o.o.} = C_1 \cdot \cos(\sqrt{6}t) + C_2 \cdot \sin(\sqrt{6}t)$$

$$y_{o.o.} = -\frac{2}{5}x - \frac{1}{5}\dot{x} =$$

$$= -\frac{2}{5}\left(C_1 \cdot \cos(\sqrt{6}t) + C_2 \sin(\sqrt{6}t)\right) - \frac{1}{5} \times$$

$$\times\left(\sqrt{6}C_2 \cos(\sqrt{6}t) - \sqrt{6}C_1 \sin(\sqrt{6}t)\right) =$$

$$= -\frac{2}{5}C_1 \cos(\sqrt{6}t) - \frac{2}{5}C_2 \sin(\sqrt{6}t) - \frac{\sqrt{6}}{5}(C_2 \cos(\sqrt{6}t) +$$

$$+ \frac{\sqrt{6}}{5}C_1 \sin(\sqrt{6}t) = \left(-\frac{2}{5}C_1 - \frac{\sqrt{6}}{5}C_2\right)\cos(\sqrt{6}t) +$$

$$+ \left(\frac{\sqrt{6}}{5}C_1 - \frac{2}{5}C_2\right) \cdot \sin(\sqrt{6}t)$$

$$\begin{cases} x_{o.o.} = C_1 \cdot \cos(\sqrt{6}t) + C_2 \cdot \sin(\sqrt{6}t) \\ y_{o.o.} = \left(-\frac{2}{5}C_1 - \frac{\sqrt{6}}{5}C_2\right)\cos(\sqrt{6}t) + \left(\frac{\sqrt{6}}{5}C_1 - \frac{2}{5}C_2\right) \cdot \sin(\sqrt{6}t) \end{cases}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{o.o.} = C_1 \cdot \begin{pmatrix} \cos(\sqrt{6}t) \\ -2\cos(\sqrt{6}t) + \frac{\sqrt{6}}{5}\sin(\sqrt{6}t) \end{pmatrix} +$$

$$+ C_2 \begin{pmatrix} \sin(\sqrt{6}t) \\ -\frac{\sqrt{6}}{5}\cos(\sqrt{6}t) - \frac{2}{5}\sin(\sqrt{6}t) \end{pmatrix}$$

Задача 11.

11. 1. $\begin{cases} \dot{x} = 3x - y + 5e^{t} \\ \dot{y} = 4x - y \end{cases}$

$\overline{X}_{ч.н.}$ в общем виде - ?

Соотв. СОЛДУ:

$$\begin{cases} \dot{x} = 3x - y \\ \dot{y} = 4x - y \end{cases}$$

$$\begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}$$

$$\begin{vmatrix} 3-\lambda & -1 \\ 4 & -1-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-\lambda & -1 \\ 4 & -1-\lambda \end{vmatrix} = (3-\lambda)(-1-\lambda) + 4 =$$

$$= -3 + \lambda - 3\lambda + \lambda^2 + 4 = \lambda^2 - 2\lambda + 1$$

$$\lambda^2 - 2\lambda + 1 = 0$$

$$\lambda_{1,2} = 1$$

Возвр. к СЛНДУ:

$$\overline{F}(t) = \begin{pmatrix} 5e^{t} \\ 0 \end{pmatrix} = e^{1\cdot t}\begin{pmatrix} 5\cos 0t + 1\sin 0t \\ 0\cos 0t + 0\sin 0t \end{pmatrix}$$

$\alpha = 1$, $\beta = 0$, $P_1(t) = 5$, $Q_1(t) = 1$, $P_2(t) = 0$, $Q_2(t) = 0$

$\alpha + \beta i = 1 \Rightarrow S = 2$

$\deg R_i = \deg T_i = \max \{ \deg(\overset{=0}{P_1}), \deg(\overset{=0}{P_2}), \deg(\overset{=-\infty}{Q_1}),$
$\deg(\overset{=-\infty}{Q_2}) \} + S = 2$

$R_1(t) = A + Bt + Ct^2$

$T_1(t) = D + Ft + Gt^2$

$R_2(t) = H + Jt + Kt^2$

$T_2(t) = L + Mt + Nt^2$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{u.u.} = e^{\alpha t} \begin{pmatrix} R_1(t)\cos(\beta t) + T_1(t)\sin(\beta t) \\ R_2(t)\cos(\beta t) + T_2(t)\sin(\beta t) \end{pmatrix} =$$
$$= e^{t} \begin{pmatrix} R_1(t) \\ R_2(t) \end{pmatrix} = e^{t} \begin{pmatrix} A + Bt + Ct^2 \\ H + Jt + Kt^2 \end{pmatrix}$$

11.2. $\begin{cases} \dot{x} = x + y + 4e^{2t}\cos t \\ \dot{y} = 3y - 2x - 5e^{2t}\sin t \end{cases}$

Coomb. СОЛДУ:

$\begin{cases} \dot{x} = x + y \\ \dot{y} = 3y - 2x \end{cases}$

$\begin{pmatrix} 1 & 1 \\ -2 & 3 \end{pmatrix}$

$\begin{vmatrix} 1-\lambda & 1 \\ -2 & 3-\lambda \end{vmatrix} = 0$

$$\begin{vmatrix} 1-\lambda & 1 \\ -2 & 3-\lambda \end{vmatrix} = 3-3\lambda-\lambda+\lambda^2+2 = \lambda^2-4\lambda+5$$

$$\lambda^2-4\lambda+5=0$$
$$D = 16-20=-4$$
$$\lambda_{1,2} = \frac{4 \pm 2i}{2} = 2 \pm i$$

Возвр. к СНЛДУ:

$$\overline{F}(t) = \begin{pmatrix} 4e^{2t}\cos t \\ -5e^{2t}\sin t \end{pmatrix} = e^{2\cdot t}\cdot\begin{pmatrix} 4\cos 1t + 0\sin 1t \\ 0\cos 1t - 5\sin 1t \end{pmatrix}$$

$\alpha=2$, $\beta=1$, $P_1(t)=4$, $P_2(t)=0$, $Q_1(t)=0$, $Q_2(t)=-5$

$\alpha+\beta i = 2+i \Rightarrow S=1$

$$\deg(R_i) = \deg(T_i) = \max\{\overset{=0}{\deg P_1}, \overset{=-\infty}{\deg Q_1}, \overset{=-\infty}{\deg P_2}, \overset{=0}{\deg Q_2}\}+$$
$$+S = 1$$

$R_1(t)=A+Bt$, $T_1(t)=C+Dt$, $R_2(t)=F+Gt$, $T_2(t)=H+Jt$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{ч.н.} = e^{2t}\begin{pmatrix} R_1(t)\cos(\beta t) + T_1(t)\sin(\beta t) \\ R_2(t)\cos(\beta t) + T_2(t)\sin(\beta t) \end{pmatrix} =$$
$$= e^{2t}\begin{pmatrix} (A+Bt)\cos t + (C+Dt)\sin t \\ (F+Gt)\cos t + (H+Jt)\sin t \end{pmatrix}$$

11.3.

$$\begin{cases} \dot{x} = 2x-3y \\ \dot{y} = x-2y+2\sin t \end{cases}$$

Соотв. СОЛДУ:

$$\begin{cases} \dot{x} = 2x - 3y \\ \dot{y} = x - 2y \end{cases}$$

$$\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}$$

$$\begin{vmatrix} 2-\lambda & -3 \\ 1 & -2-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2-\lambda & -3 \\ 1 & -2-\lambda \end{vmatrix} = -4 - \cancel{2\lambda} + \cancel{2\lambda} + \lambda^2 + 3 = \lambda^2 - 1$$

$$\lambda^2 - 1 = 0$$

$$\lambda_{1,2} = 1$$

Возвн. к СЛАДУ:

$$\bar{F}(t) = \begin{pmatrix} 0 \\ 2\sin t \end{pmatrix} = e^{0 \cdot t} \cdot \begin{pmatrix} 0\cos 1t + 0\sin 1t \\ 0\cos t + 2\sin t \end{pmatrix}$$

$$\alpha = 0,\ \beta = 1,\ P_1(t) = 0,\ Q_1(t) = 0,\ P_2(t) = 0,\ Q_2(t) = 2$$

$$\alpha + \beta i = i \Rightarrow S = 0$$

$$\deg(R_i) = \deg(T_i) = \max\{\underbrace{\deg(P_1)}_{=-\infty}, \underbrace{\deg(P_2)}_{=-\infty}, \underbrace{\deg(Q_1)}_{=-\infty}, \underbrace{\deg(Q_2)}_{=0}\} + S = 0$$

$$R_1(t) = A,\ R_2(t) = C,\ T_1(t) = B,\ T_2(t) = D$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{u.u.} = e^{ot} \begin{pmatrix} A\cos t + B\sin t \\ C\cos t + D\sin t \end{pmatrix}$$